Course
« *Deep Learning*»

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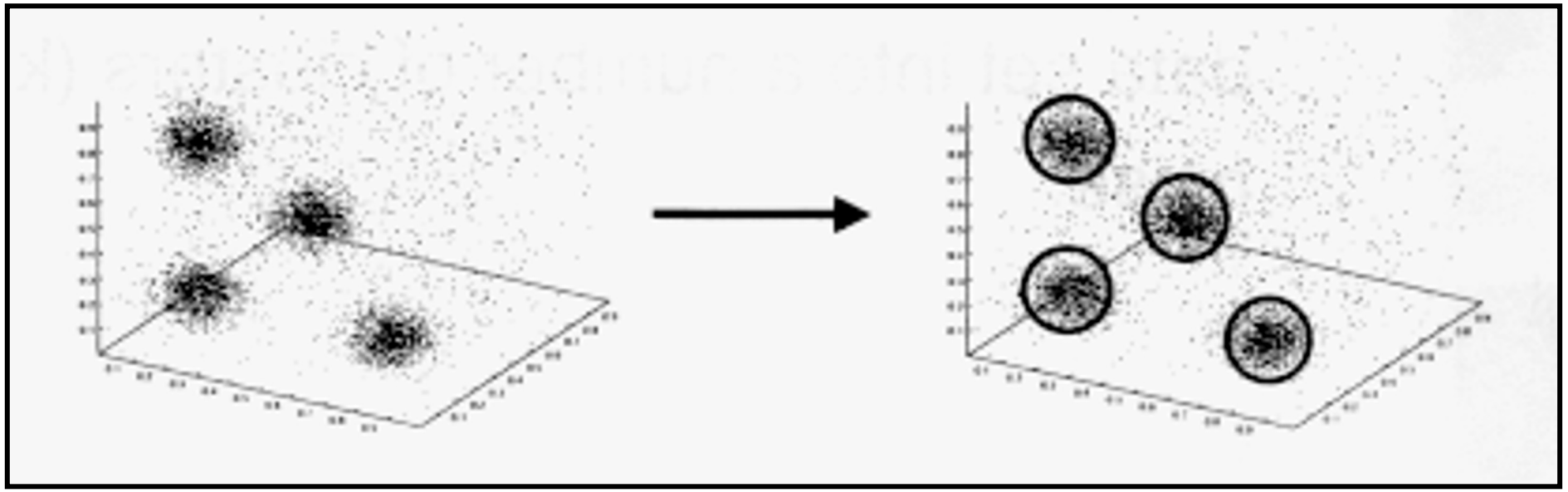
ENSIA

2026-2027

3. *Clustering (unsupervised classification)*

Objective

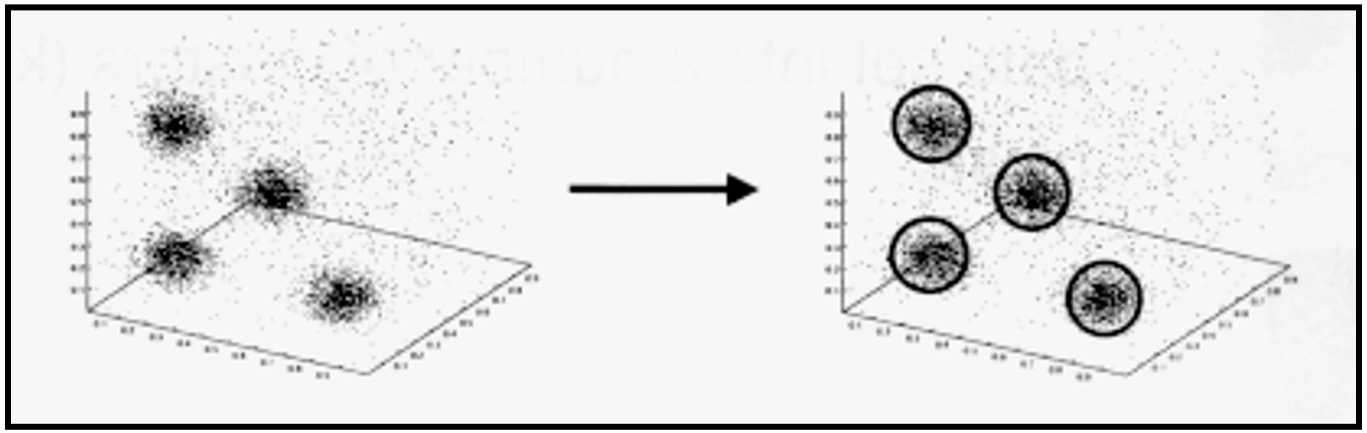
- Automatically partition a dataset using a similarity measure/distance.



3. *Clustering (unsupervised classification)*

Objective

- Input:
  - A set of *N* vectors describing *N* individuals (also called *samples, observations…).*
  - A set of parameters (related to the clustering algorithm).
- Output:
  - *k* clusters
  - + noise (possibly).



3. *Clustering (unsupervised classification)*

Applications

- Summarize large datasets
  - A set of individuals described using *d* variables (features).

Objective

- Put together similar individuals
- Instead of studying all of the individuals:
  - ⇒ Simply studying the characteristics of the group rather than studying each of the individuals who compose it.
  - ⇒ Study the average individual in each group.

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Applications

- Image segmentation

Input:

- Pixels of an image (values, position…).

Sortie :

Groups of similar and connected pixels. *Homogeneous region = cluster of pixels*





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Applications

- **Bioinformatics:** grouping animals according to their biological features to reconstruct phylogeny trees
- **Business:** dividing customers into segments or forming a hierarchy of employees based on salary.
- **Image processing:** grouping handwritten characters in text recognition based on the similarity of the character shapes.
- **Information retrieval:** categorizing search results based on the query.
- …

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In practice: 4 steps

1. Description of the individuals
    - Feature extraction of individuals.
    - Feature selection with respect to the application.
2. Definition of a similarity measure between individuals (a distance, for example).
3. Grouping individuals using their feature vectors and the associated similarity measure.
4. Making use of clusters: Interpretation, abstraction…

3. *Clustering (unsupervised classification)*

Existing techniques: An overview

Data partitioning	**Hierarchical**	**Density-based**	**Space partitioning**
• K-means • CLARANS • Kohonen maps	• Traditional methods • BIRCH	• DBSCAN • DENCLUE	• STING • Wave Cluster • Opti-Grid
• Iterative algorithms that refines an initial clustering.	• Hierarchical partitioning of the data.	• Based on density estimation methods (parametric and non-parametric)	• Space quantization.

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Quality criterion

- The spread of each cluster.

  $\Rightarrow$ intra-cluster variance

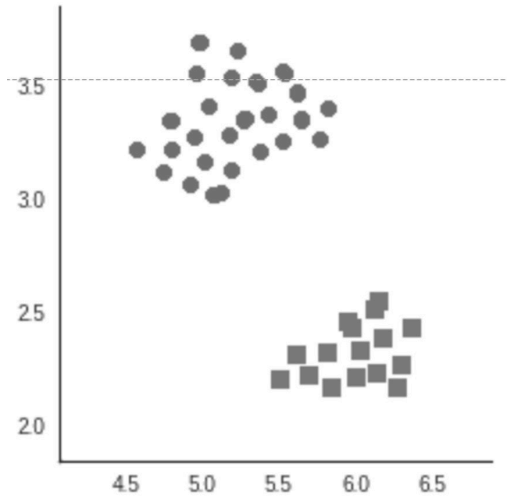
n: total number of individuals *(v)*.

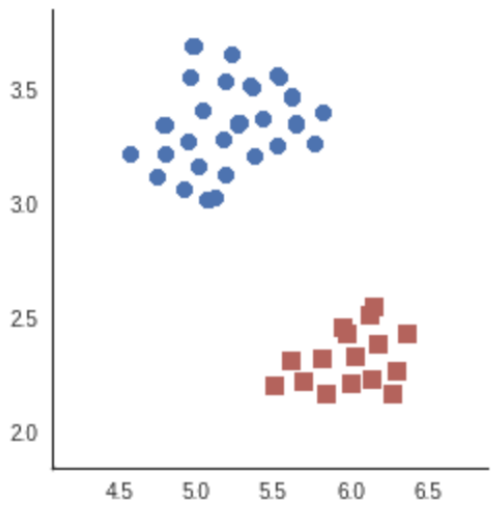
k: number of clusters.

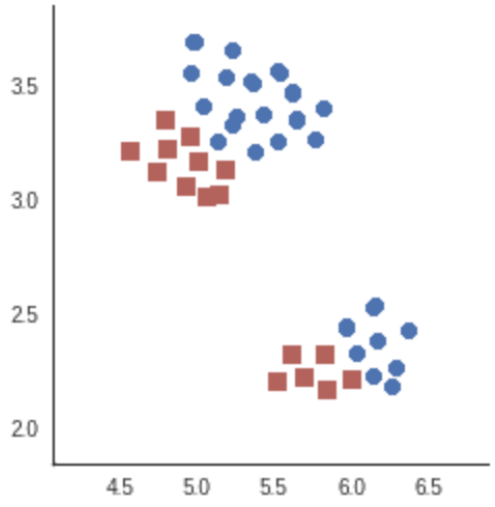
n_j: number of individuals in clusters j

g_j : centroid of cluster j

$$Intra\ cluster\ variance = \frac{1}{n}\sum_{j=1}^{k}\sum_{i=1}^{n_j} d(v_i, g_j)^2$$







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Quality criterion

- Spacing of clusters from each other.
- If we consider only centroids:
  - ⇒ Study the spread of centroids.
  - ⇒ Inter-cluster variance.

$$Inter\ cluster\ variance = \sum_{j=1}^{k} \frac{n_j}{n} d(g, g_j)^2$$

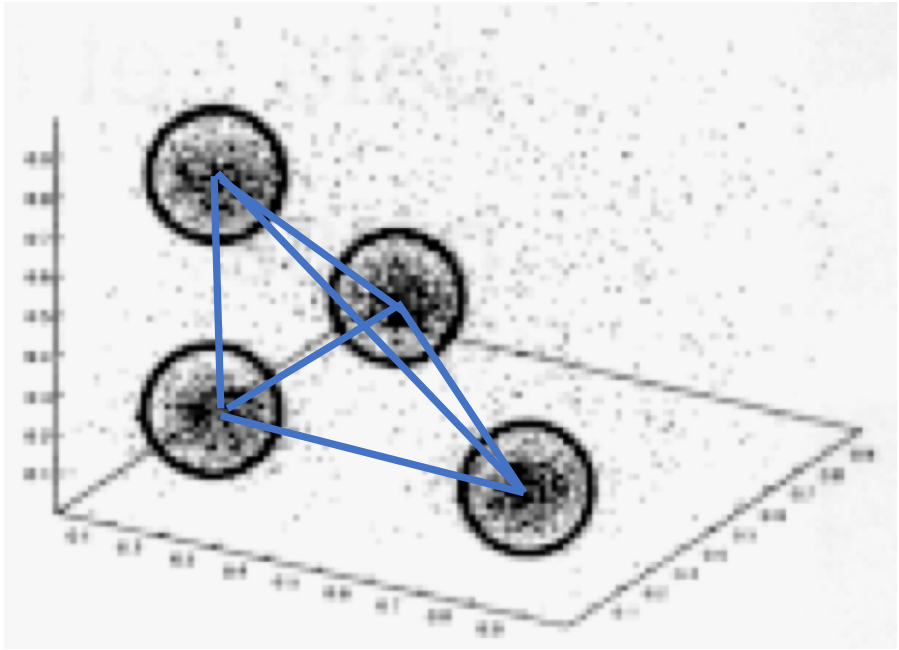
n: total number of individuals.

k: number of clusters.

n_j: number of individuals in clusters j

g_j: centroid of cluster j

g: centroid of the dataset.



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Quality criterion

- We seek a clustering in which clusters are:
  1. Tightened on themselves => intra-cluster variance low.
  2. Far from each other => inter-cluster variance high.
- The goal is to minimize the intra-cluster variance and to maximize inter-cluster variance**.**
- However, the two objectives are equivalent => All you have to do is to optimize one of the two criteria.
- **Total variance = intra-cluster variance + inter-cluster variance**
- The total variance is constant.

(Theorem of Huygens).

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K-Means: The algorithm

Input:

N individuals.
k centers *[randomly chosen at the beginning].*
k empty clusters.

Algorithm:

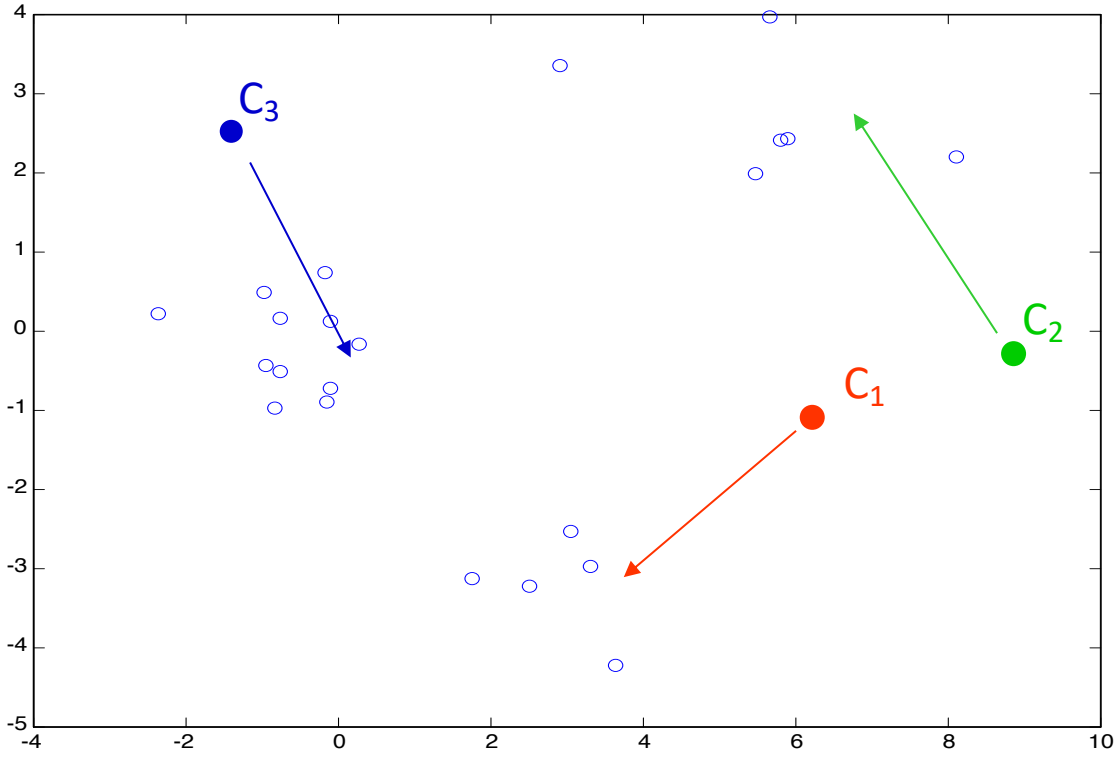
Repeat:

- Distribute the *N* individuals over the *k* clusters using the rule of « the nearest center ».
- Compute the new k centers.

Until two consecutive “clusterings” are identical

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K-Means: “perform clustering around moving centers”



3. *Clustering (unsupervised classification)*

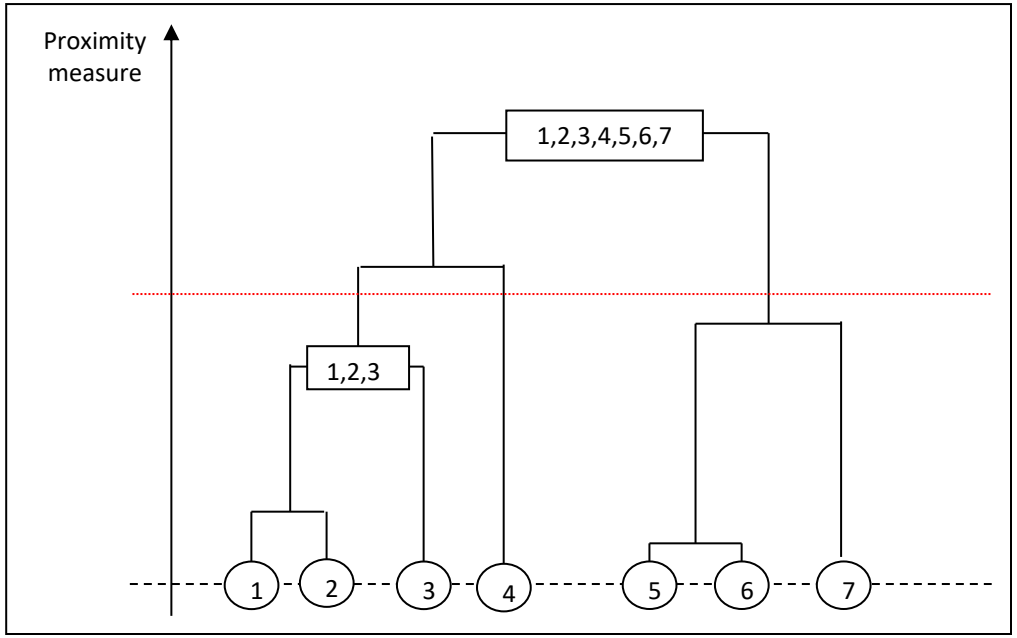
K-Means: The algorithm

- Parameters: The number of clusters
- The algorithm is sensitive to the initial conditions (non-deterministic).
- Iterative algorithm that requires several passes over the dataset.
- Complexity: O(N).

3. *Clustering (unsupervised classification)*

Hierarchical clustering

- Clusters are visually represented in a hierarchical tree called a ***dendrogram***.
- There is no need to pre-specify the number of clusters. Instead, the dendrogram can be cut at the appropriate level to obtain the desired number of clusters.
- Data is easily summarized/organized into a hierarchy using dendrograms. Dendrograms make it easy to examine and interpret clusters.
- Requires a proximity measure between clusters.



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Hierarchical clustering

There are two main types of hierarchical clustering:

1. **Agglomerative**: Initially, each object is considered to be its own cluster. According to a particular procedure, the clusters are then merged step by step until a single cluster remains. At the end of the cluster merging process, a cluster containing all the elements will be formed.

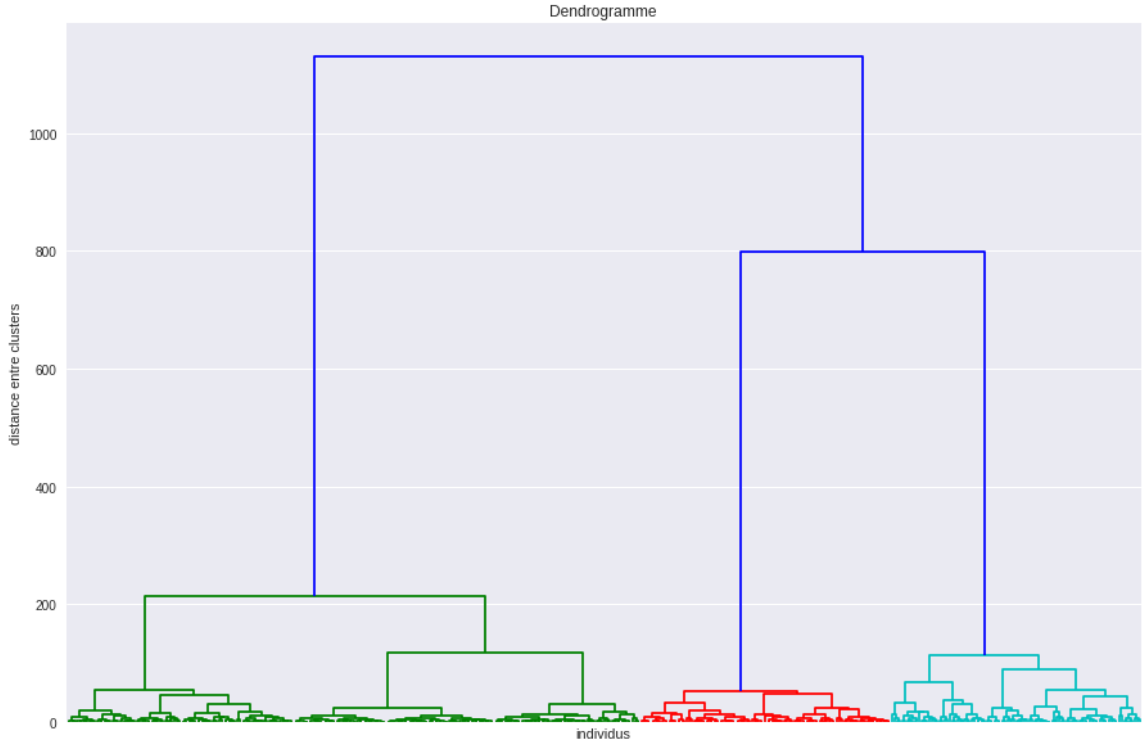
2. **Divisive**: Initially, all objects are considered in a single cluster. Then the division process is performed step by step until each object forms a different cluster.

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Hierarchical clustering: Graphical representation

If we have a very large dataset, the bottom of the dendrogram may be unreadable.

$\Rightarrow$ Solution: Show and focus on the top part of the tree.

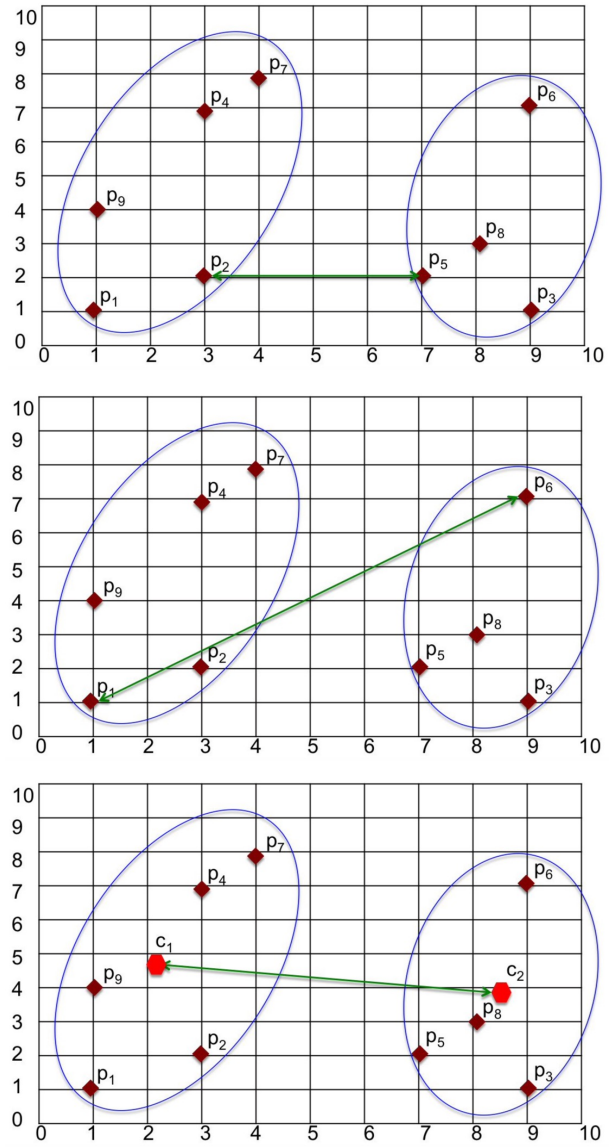


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Hierarchical clustering

How to compute the proximity (similarity) between two clusters?

- **Min (Single) Linkage:** The distance between 2 clusters is the minimum distance between points in those clusters.
- **Max (Complete) Linkage:** The distance between 2 clusters is the maximum distance between points in those clusters.
- **Average Linkage:** The distance between 2 clusters as the average pairwise distance among all pairs of points in the clusters.
- **Centroid Linkage:** The distance between 2 clusters is the distance between their centroids.



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Hierarchical clustering

How to compute the proximity (similarity) between two clusters?

⇒ **Ward Linkage**

The Ward approach analyzes the variance of the clusters rather than measuring distances directly, minimizing the variance between clusters.

It consists of grouping clusters so that the increase in inter-cluster variance is maximum.

This amounts to the same thing according to Huygens' theorem: The increase in intra-class inertia is minimum.

Ward's method is preferred more than others in clustering.

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Hierarchical clustering

Properties of the algorithm

- Unlike k-means, hierarchical classification does not require choosing a number of clusters beforehand.
- Algorithm independent of the order of insertion of points.
- Complexity: $O(n^3)$.